

NEHA SINGH

Plot no. 60 Pragati Path , Chattrashal Nagar , Malviya Nagar , Jaipur , 302017
ne2003singh@gmail.com | 797-6682-829

SUMMARY

I'm a creative technologist who can debug a pose estimation model and design a poster without spilling my coffee. With 5 months building AI-powered tools at Day Dream Studio, I've learned that the best code is the kind that actually ships—whether that's 20+ custom VFX nodes, a complete desktop app for dataset captioning, or comprehensive docs that people might actually read. I work in that sweet spot where "make it work" meets "make it not ugly," using Python, React, Flask, and whatever AI tools get the job done. Currently looking for remote roles where I can build things that matter, learn fast, and maybe occasionally use a kaleidoscope effect node.

TECHNICAL SKILLS

- **Technical:** Python, Flask, React, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, Node.js, SQL, Git, NumPy, OpenCV, Pillow, PyTorch, TensorFlow, scikit-learn
- **AI/ML & Computer Vision:** Claude API, ChatGPT, Gemini, Grok, Moondream, YOLOX, DWPose, MediaPipe, Vision Transformers (ViT), Hugging Face Transformers, ComfyUI
- **API Tools & Frameworks:** Librosa (audio), FFmpeg (video), LiteGraph, shadcn/ui, Docker, RunPod, PySide6/PyQt, PyInstaller
- **Design Tools:** Adobe Photoshop, Adobe Express, Canva, Inkscape, Figma, Blender (beginner), Notion
- **Other:** Prompt Engineering, UI/UX Basics, Technical Documentation, Microsoft Office Suite
- **Soft Skills:** Creative thinking, problem-solving, adaptability, teamwork, communication, independent learning

RELEVANT EXPERIENCE

AI Creative Technologist Intern - Day Dream Studio

September 2025 - March 2026

Led technical implementation and documentation for multiple AI-powered creative tools, working across full-stack development, computer vision integration, and comprehensive QA workflows.

Node-Based VFX System (Blobit) - Backend Development & Documentation

- Implemented 20+ custom processing nodes including YOLOX object detection, DWPose pose estimation, SAM2 segmentation, motion blur, HSL masking, and various image/video filters; debugged AI model integration issues (MediaPipe, PyTorch/ONNX conflicts)
- Authored comprehensive technical documentation for 100+ nodes using Astro/Starlight/MDX with parameter tables, usage examples, and best practices
- Set up Python ML infrastructure (PyTorch, ONNX Runtime, OpenCV, Librosa), implemented audio-reactive node system with real-time spectral analysis, and standardized array synchronization logic across nodes.
- **Tech Stack:** Python, Flask, PyTorch, ONNX Runtime, MediaPipe, YOLOX, NumPy, OpenCV, Librosa, Astro, MDX, Git

AI Mockup Generation Platform (Mockup Studio) - Full-Stack Development

- Contributed to full-stack development of AI-powered mockup generation tool using React/TypeScript frontend and Flask backend, implementing product-type detection with Vision Transformers (ViT)
- Integrated ComfyUI workflow execution pipeline (image upload → node updates → queuing → output retrieval), built preset management system for reusable prompt configurations, and developed download/batch processing features
- Worked across React UI components and Flask API endpoints under senior mentorship while debugging workflow integration issues and optimizing generation pipeline performance
- **Tech Stack:** React, TypeScript, Vite, Tailwind CSS, shadcn/ui, Python, Flask, PyTorch, Hugging Face Transformers, ComfyUI API, Git

Dataset Preparation Tool - Desktop Application Development

- Redesigned and expanded PySide6 desktop application for AI-powered image captioning, implementing integration with 7 VLM backends (Claude, GPT, Gemini, Grok, Moondream variants) and building dynamic prompt engineering system with model-specific formatters (SD 1.5/XL, WAN, Flux, QWEN)
- Redesigned UI from vertical to tabbed interface, developed batch processing with auto-save and real-time caption editing, created CSV export for training datasets, and implemented comprehensive error handling
- **Tech Stack:** Python, PySide6/PyQt, Claude API, OpenAI API, Gemini API, Moondream, NumPy, Pillow, Git

Cloud Rendering Platform (BananaFarm) - Quality Assurance & Testing

- Conducted comprehensive end-to-end testing for cloud-based Blender rendering service across GPU tiers (RTX 3090/4090), performing 100+ test renders with systematic performance benchmarking, and created detailed test matrices for still/animation renders with different engines (Cycles/Eevee) and devices (CUDA/OptiX)
- Identified and documented critical bugs (hardware selection defaults, VRAM spikes, codec errors), compiled test reports with time/cost breakdowns and comparison charts, and provided UX/UI feedback that improved onboarding flows
- **Tech Stack:** Blender 4.5/5.0, RunPod GPU infrastructure, FFmpeg, Docker

Freelance Graphic Designer - Stardust Observatory **March 2025 - April 2025**

- Designed branding and promotional materials for a public astronomy initiative near Ranthambore that brings stargazing and science to the public.
- Created visual assets for outreach events, astrophotography promotions, and stargazing sessions.

Freelance Graphic Designer -Rajasthan University Women’s Association (RUWA) **November 2024 - January 2025**

- Designed official posters, invitations, and visual materials for the RUWA Golden Jubilee International Conference.
- Delivered high-quality designs aligned with the event’s theme and academic prestige.

KEY PROJECTS

WellCheck – AI-Powered Health App **May 2025- June 2025**

- Built a web app that collects vitals (BP, SpO2, pulse, temperature, calories) and gives personalized health suggestions using ML models (scikit-learn).

Pythonia – AI-Powered Python Learning Game **December 2024 - March 2025**

- Created a gamified platform using AI-driven multiple-choice prompts to teach Python basics interactively.
- Integrated scoring, progression mechanics, and story elements to boost retention and engagement.

EDUCATION

Kanoria P G Mahila Mahavidyalaya , Jaipur

Bachelor of Computer Application

GPA: 3.05 / 4.00

KEY COMPETENCIES

Full-Stack Development • AI/ML Integration • Computer Vision • Technical Documentation • Problem-Solving • Quality Assurance • Independent Learning • Collaboration • Creative-Technical Hybrid Projects

ACHIEVEMENTS

- **Google Generative AI Learning Path** – Certification in AI and prompt engineering.
- **Intro to Graphic Design with Photoshop** – Great Learning.
- **Organizing Committee Member** – College Techfest "TECHKSHITIJ" (2 years, 2023-2024).
- **Lead designer for annual college Techfest brochures (2022-2024, 3 consecutive years).**
- Active participant in inter-college fests, hackathons, and seminars on AI, full-stack development, and cybersecurity.